Predicting Falcon 9 First-Stage Landing Success

Applied Data Science Capstone • IBM Professional Certificate

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https://github.com/IgorGryb/applied-data-science-capstone

Executive Summary

Business question: will the Falcon 9 first stage land successfully? SpaceX advertises ~USD 62M per launch vs ~USD 165M+ for competitors — the gap is first-stage reuse.

Data: 90 Falcon 9 launches (Falcon 1 excluded) from the SpaceX API v4 + Wikipedia landing records; first-stage landing success rate 67%.

Methods: SQL analytics, seaborn/matplotlib EDA, Folium geospatial analysis, Plotly Dash interactive dashboard, and classification with GridSearchCV.

Result: a Decision Tree predicts landing success with 83.33% test accuracy (15 of 18 unseen launches) — the best of four tuned models.

Introduction

SpaceX's cost advantage comes from recovering and re-flying the Falcon 9 first stage.

If we can predict landing success from launch parameters (payload, orbit, site, booster block), we can estimate the true expected cost of a launch before it flies.

This project follows the full data-science lifecycle: collect → wrangle → explore → visualise → predict, on 90 real Falcon 9 missions.

Historical anchor: SpaceX's first static fire test was conducted in 2006; routine booster recovery began a decade later.

Data Collection Methodology

Source 1 — SpaceX REST API v4: POST /launches/query filtered to Falcon 9; joined rocket, launchpad, payload and core documents (see notebook 1).

Source 2 — Wikipedia 'List of Falcon 9 and Falcon Heavy launches': scraped with requests + BeautifulSoup for the booster-landing outcome column the API lacks.

Normalised to one row per launch: FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Customer, coordinates.

Result: data/dataset_spacex.csv — 90 launches, 2010–2023, zero Falcon 1 rows.

Data Wrangling Methodology

Type checks and missing-value audit; payload mass clipped to the physical range 200–15,600 kg.

Target engineering: Class = 1 when Outcome ∈ {True RTLS, True ASDS, True Ocean}, else 0.

Verified landing success rate: 60/90 = 66.7% ≈ 67%.

One-hot encoding of 14 categorical attributes → 80 binary columns; plus 3 numeric columns (FlightNumber, PayloadMass, ReusedCount) → 90 × 83 feature matrix.

EDA & Interactive Visual Analytics Methodology

SQL (SQLite table SPACEXTBL): unique sites, payload totals/averages, first successful landing date, success rate by orbit and by site.

Visual EDA: payload-vs-flight scatter, success-rate bars by orbit/site, yearly cadence trend.

Geospatial: Folium map with launch-site circles sized by launch count and popups with per-site success rates.

Interactive dashboard: Plotly Dash app — site dropdown, success pie chart, payload-range slider, payload-vs-success scatter coloured by booster version.

Predictive Analysis Methodology

train_test_split(test_size=0.2, random_state=2) → 72 train / 18 test records.

Four classifiers, each tuned with GridSearchCV (10-fold CV, accuracy scoring):

• Logistic Regression — C ∈ {0.01, 0.1, 1}; • SVM — kernel ∈ {linear, rbf, poly, sigmoid}, C ∈ {0.1, 1, 10}, gamma ∈ {scale, auto};

• Decision Tree — criterion, max_depth, min_samples_split; • KNN — k ∈ 1..10, weights, p.

Evaluation on the held-out 18 launches: accuracy + confusion matrices; best model selected by test accuracy.

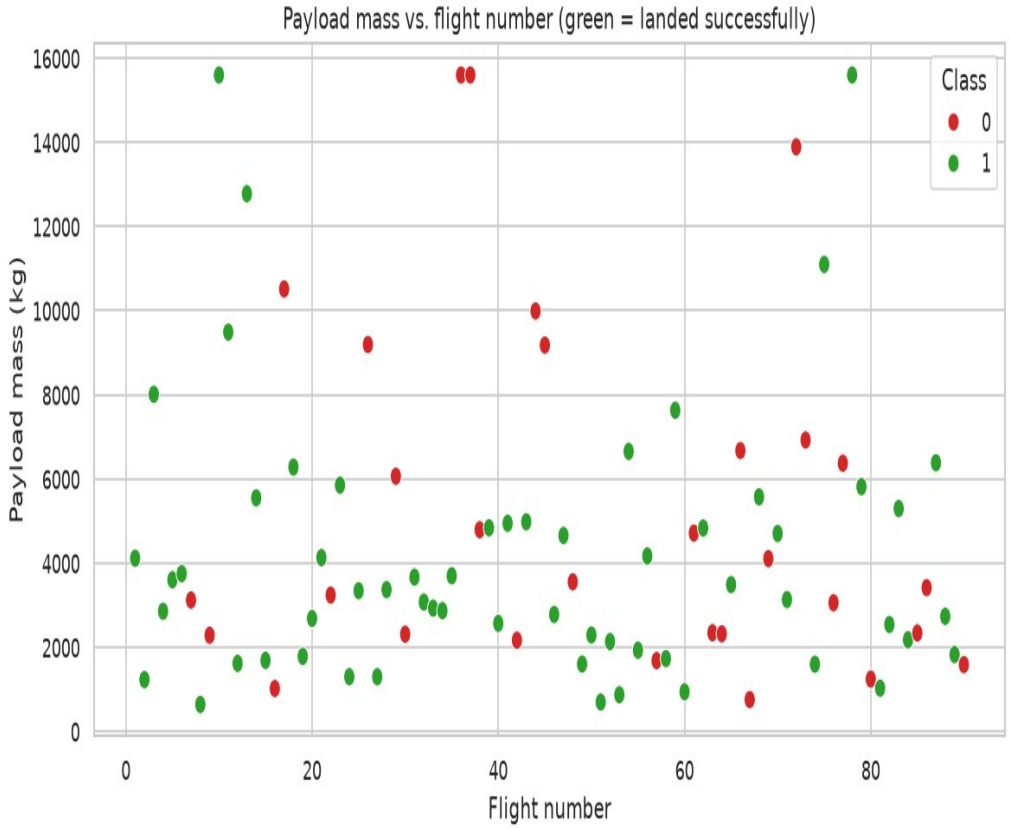
EDA Results — Visual Analysis

Heavier payloads and high-energy orbits (GTO/GEO) show markedly lower recovery rates.

Success rate climbs steeply after 2016 as landing procedures mature (Block 5 era).

LEO / ISS / VLEO missions recover boosters far more often than deep-space missions.

Florida sites dominate cadence; VAFB serves polar/SSO missions.



Payload mass vs. flight number (green = landed)

EDA Results — SQL Analysis

Metric	Result
Total Falcon 9 launches	90
Unique launch sites	4 (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E)
Launches per site	LC-40: 23 SLC-40: 30 LC-39A: 22 SLC-4E: 15
Landing success rate (overall)	66.7% (60 of 90)
Best orbit for recovery	LEO / ISS / VLEO missions
Hardest orbit for recovery	GTO / GEO missions

All metrics computed with SQL on SPACEXTBL (see notebook 3). Site ranking by recovery success directly informs launch-site selection.

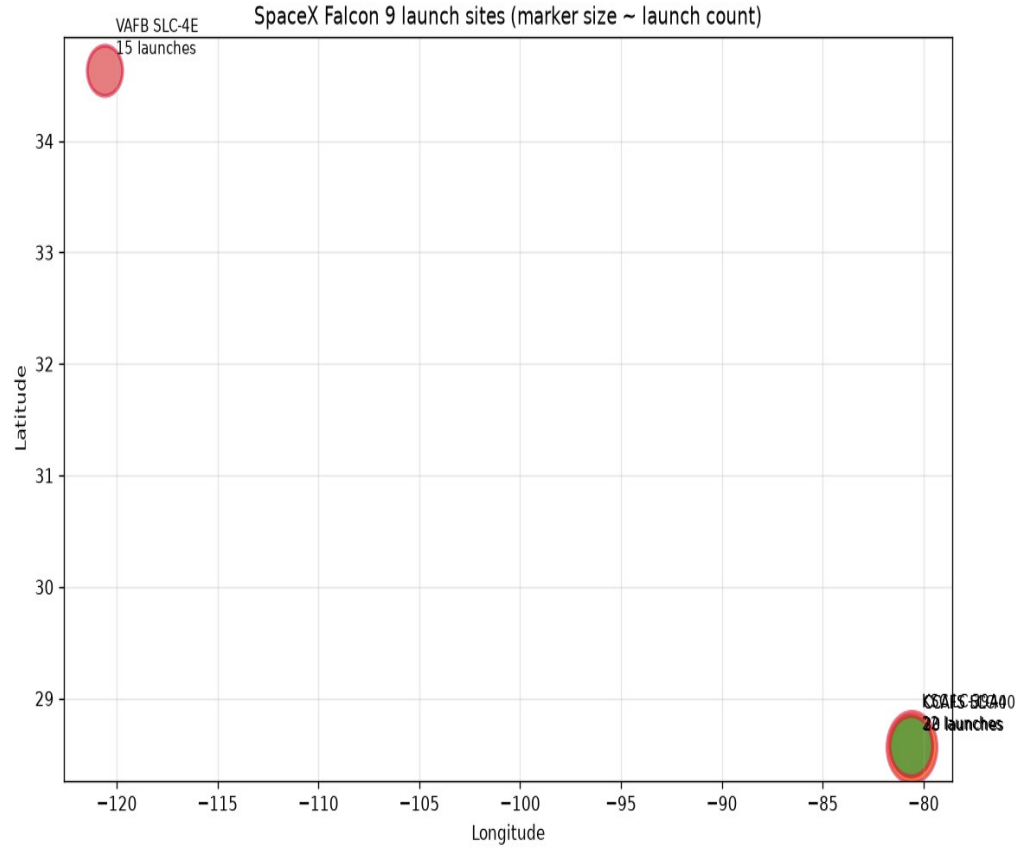
Folium Map Results

Three Florida pads (LC-40, SLC-40, LC-39A) handle ~83% of Falcon 9 launches.

VAFB SLC-4E (California) is the polar/SSO gateway.

RTLS ground landings cluster at LZ-1/LZ-2/LZ-4 next to the Florida pads; ASDS drone-ship recoveries (OCISLY, JRTI, ASOG) serve higher-energy trajectories.

Interactive map with per-site popups: assets/launch_sites_map.html.



Launch sites (marker size ~ launch count)

Plotly Dash Dashboard Results

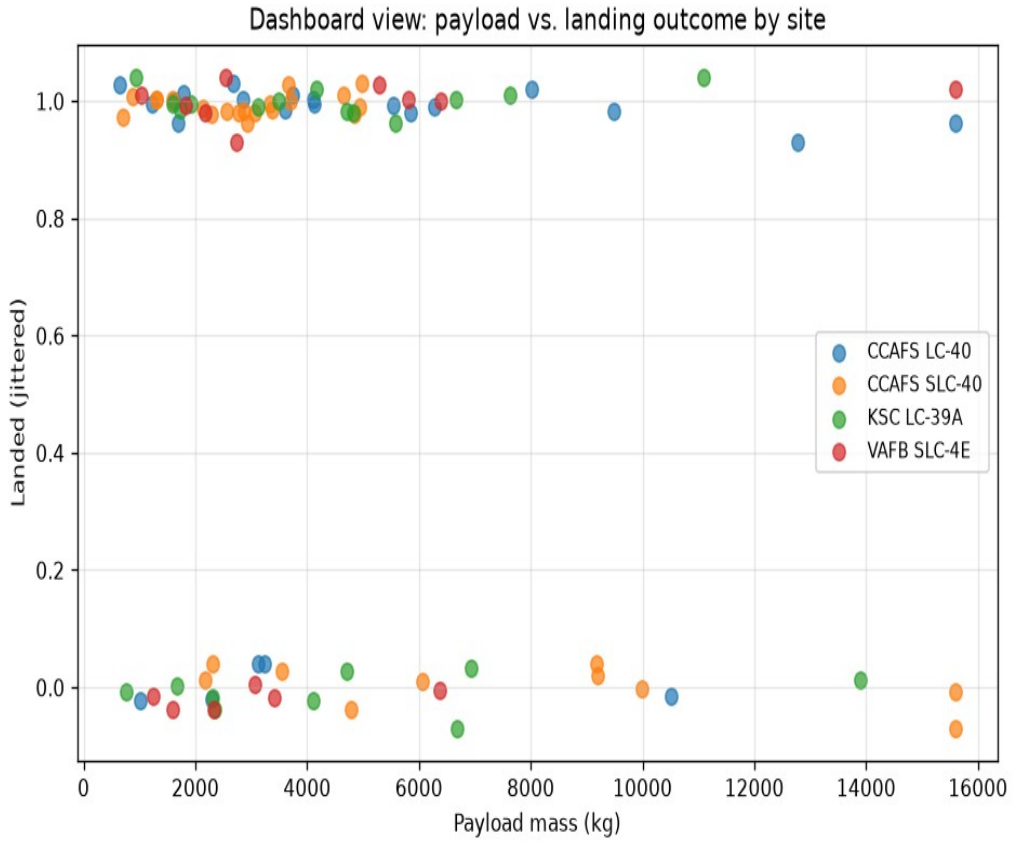
Site dropdown + success pie chart: instant success/failure share per pad.

Payload-range slider (200–15,600 kg): filters the scatter live.

Scatter (payload vs. outcome, coloured by booster version): Block 5 + light payloads → dense green cluster; heavy GTO payloads → red.

Key interaction finding: narrowing the slider to < 8,000 kg lifts the visible success share dramatically — the dashboard makes the payload effect tangible.

Runnable app: notebooks/spacex_dash_app.py → http://127.0.0.1:8050.



Static preview of the dashboard scatter (interactive in the app)

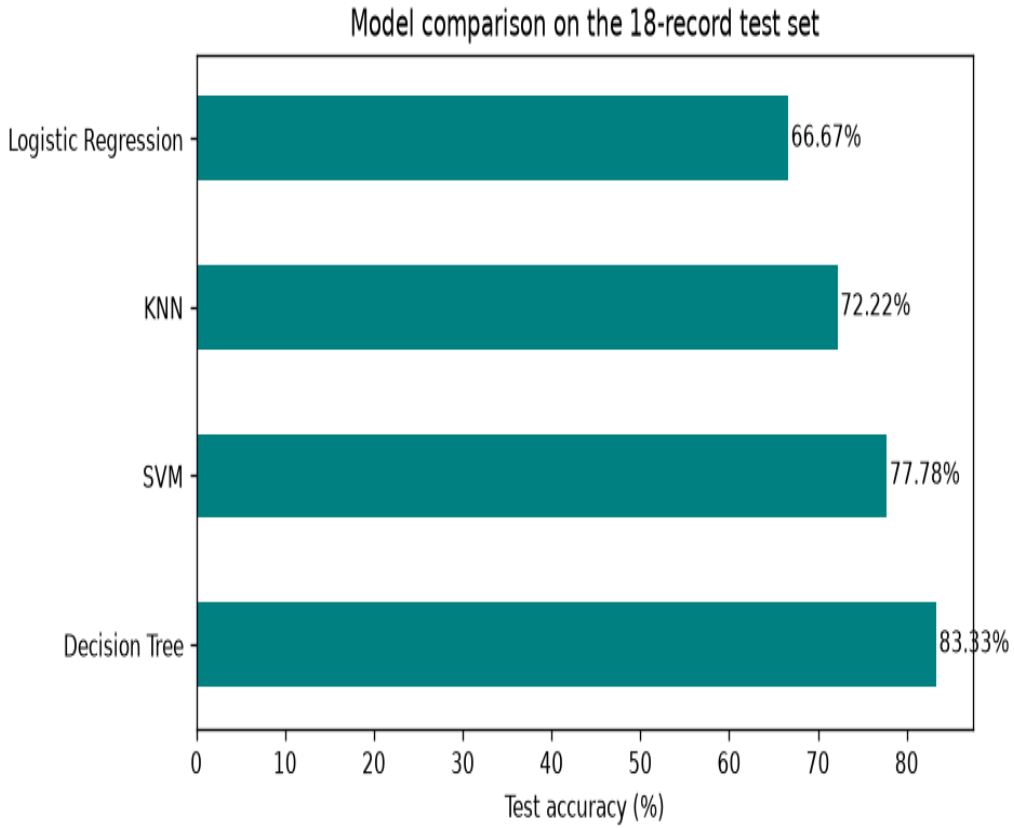
Predictive Analysis — Classification Results

Decision Tree: 83.33% test accuracy (15/18) — best model.

SVM: 77.78% — best kernel found by grid search: sigmoid.

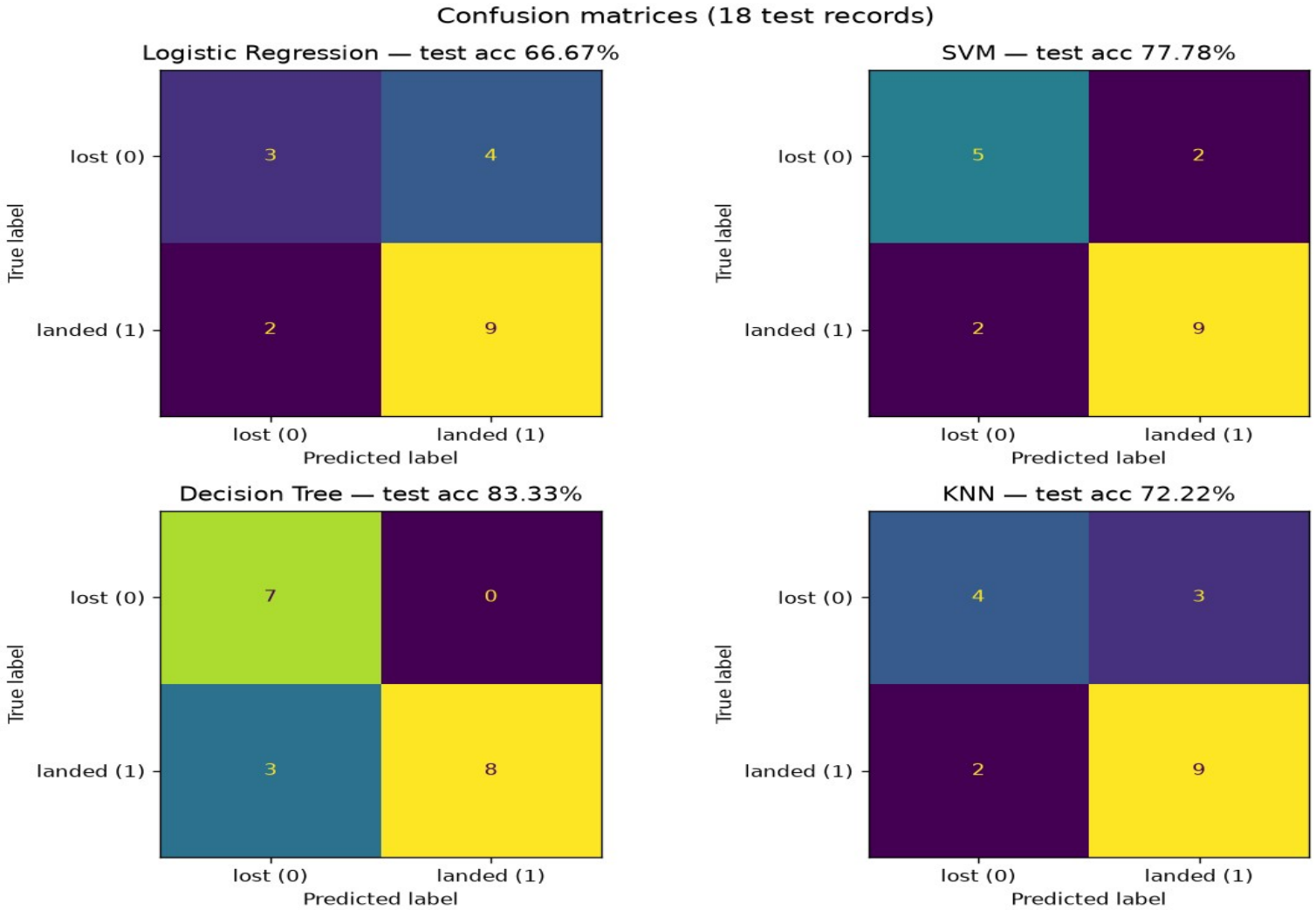
Logistic Regression: 66.67%; KNN: 72.22%.

The tree wins because landing success is driven by nonlinear interactions (payload × orbit energy × booster block), which a linear boundary cannot separate.



Test accuracy on 18 held-out launches

Confusion Matrices (Test Set, n=18)



Conclusion

All four project questions answered: data collected, wrangled, explored visually and geospatially, served through an interactive dashboard, and modelled.

The Decision Tree (83.33% test accuracy) is the recommended predictor for go/no-go recovery planning and launch-cost estimation.

Operational takeaway: keep payloads under ~8 t on Block 5 boosters from Florida pads for the highest recovery probability.

All code, data and this report: https://github.com/IgorGryb/applied-data-science-capstone

Innovative Insights

Cost model: at ~USD 100M saved per recovered booster and a 67% recovery rate, the expected reuse saving is ~USD 67M per launch — the predictor turns this into a per-mission expected-value estimate.

Booster longevity: the most-flown serials in the data are all Block 5 airframes, confirming that block upgrades — not just procedures — drove the post-2016 success-rate jump visible in the yearly trend.

Next step: add weather-at-launch and sea-state features for ASDS recoveries; the current 83.33% is limited by unobserved operational factors.